Preparation of the TFG manuscript using LaTeX

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Abstract: Some examples of the use of LaTeX, for styles, sections, figures, tables, formulas, references, and other details are provided here with specific information and guidelines that should be read carefully before starting to write the manuscript for the Treball Fi de Grau (TFG). Note that the report must be written in English, but on the sixth page, the title, summary, and keywords in Catalan must be included. The sixth page should also contain a brief comment relating the TFG to the Sustainable Development Goals (SDG). The abstract should concisely cover the objectives, methods and data used, results, and conclusion, and it should be 100 to 150 words in length.
Keywords: Up to 6 key words to correlate your work with the Physics degree's subjects
SDGs: Enumerate the sustainable-development goals (ODS) this work is related to (see page 6)

I. INTRODUCTION

In this document you will find the structure as well as the length of what the *Treball Fi de Grau* (TFG) report is required to have. It uses LaTeX in a two-column format and provides an example of what the essay ought to look like, including lists, tables and figures.

The text has to be written in (good) English (compulsory). The length of the report should by no means exceed **five** pages in the present format, all figures, tables, etc..., included. It is important that the two requirements are precisely met. Otherwise the essay will not be considered for evaluation, and will automatically fail with a mark of *Suspès*.

The report must consist of the following parts:

1. Abstract
2. Introduction
3. Development sections
4. Conclusions
5. Bibliography
6. Appendix (Optional)

The final score will combine both the advisor and the jury marks. The defense will consist of a 15 minute public presentation before a jury composed by two members. The deadlines as well as the dates for the presentations will be announced in due time. The students are encouraged to stick to the allowed allotted time (15 minutes), since no extra time will, in general, be conceded by the jury members. At any rate, it should be clear that a list of conclusions needs to be read before the presentation is over.

The evaluation will follow the *Rúbriques* (or *Taules d'avaluació*) provided by the TFG Advisory Committee, which will be available to the students prior to the presentation date, well ahead of time. It is highly advisable for the students to read them thorough and carefully, in order to become aware of all the competences that need be assessed to comply with the guidelines of Bologna.

Make sure that the presentation will include:

1. Introduction.
2. Development sections.
3. Conclusions.

You can find all the information concerning the TFG process in,

`https://www.ub.edu/portal/web/fisica/tramit-tfg`

II. EQUATIONS: A FEW EXAMPLES

Let us write a few equations,

$$\Delta = \sum_{i=1}^{N} w_i (x_i - \bar{x})^2, \tag{1}$$

with a label so that we can refer to it later. For instance you can refer to it as Eq. (1). All equations must be numbered. If needed, you can also produce unnumbered equations.

$$T = 2\pi \sqrt{\frac{l}{g}}.$$

You can insert equations in the text, like $e^{i\pi} + 1 = 0$; $\frac{df(x)}{dx}$ or you may prefer $\dfrac{df(x)}{dx}$. You can also write aligned equations,

$$\begin{aligned} \frac{du(x)}{dx} + Vu(x) + Wv(x) &= (\lambda + \alpha)\, u(x) \\ \frac{dv(x)}{dx} + Vv(x) + Wu(x) &= (\lambda - \alpha)\, v(x). \end{aligned} \tag{2}$$

Further examples are,

$$\begin{aligned} \nabla\cdot\vec{E} &= 4\pi\rho \\ \nabla\times\vec{B}-\frac{1}{c}\frac{\partial\vec{E}}{\partial t} &= \frac{4\pi}{c}\vec{J} \\ \nabla\times\vec{E}+\frac{1}{c}\frac{\partial\vec{B}}{\partial t} &= 0 \\ \nabla\cdot\vec{E} &= 0, \end{aligned} \tag{3}$$

as can be found in [3], pag. 14. Also,

$$E=-J\sum_{i=1}^{N}s_i s_{i+1}. \tag{4}$$

You can define your own macros to save typing [6]. For example, suppose that you introduce a macro as follows:

```
\newcommand{\El}{{\vec{E}(\vec{x},t)}}
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Then, by simply typing,

```
$\El$
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you get the electric field $\vec{E}(\vec{x},t)$ printed out (in math mode, of course).

You can write $\tilde{x}$, $\overline{A}$; %, accents: Schrödinger and é, à, À, Martí; calçotada, $A\equiv B$, $\vec{A}\cdot\vec{B}$, $\vec{A}\times\vec{B}$, $\vec{v}$, $\hat{n}$, $c\sim d$, $z\approx w$, $\langle x\rangle$, $\langle\psi|X|\psi\rangle$, $\hbar$, g_{ij}, δ^i_j, $R_{\mu\nu\alpha\beta;\xi}$.

You can also quote references easily, i.e., [4], [5].

A matrix

$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \tag{5}$$

a vector (column)

$$\begin{pmatrix} 1 \\ \sqrt{3} \\ \pi \end{pmatrix}, \tag{6}$$

a row,

$$\begin{pmatrix} p & q & r \end{pmatrix}, \tag{7}$$

and a determinant,

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}. \tag{8}$$

Further examples:

$$\int_{-\infty}^{\infty} e^{-x^2}\,dx=\sqrt{\pi}, \tag{9}$$

also,

$$\int_a^b f'(x)\,dx=f(b)-f(a), \tag{10}$$

and so on.

A. Tables and figures

An example of a table is shown in Table I.

TABLE I: Force, area, and pressure data for an experiment with glass syringes.

	Piston 1	Piston 2
Force (N)	4.40	2.25
Area (cm^2)	6.16	3.14
Pressure (N/cm^2)	0.714 ± 0.002	0.716

Here we try to insert an illustrating figure, Fig. 1. For that you need to have created the figure before, in this case the fig1.pdf file.

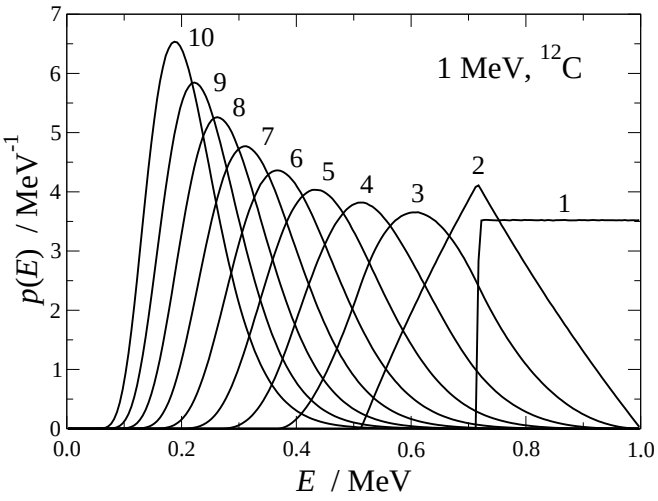


FIG. 1: A sample figure.

III. DEVELOPING SECTIONS

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TABLE II: Another table.

M1	Teoria:	Galileo	A11G
(11:45)	Problemes:	Newton	A11G
	Problemes	Planck (M1A)	A11G
(12:45)	tutoritzats:	Fermi (M1B)	A23M
M2	Teoria:	Clausius	A12G
(8:30)	Problemes:	Goeppert Mayer	A12G
	Problemes	Boltzmann (M2A)	A12G
(9:30)	tutoritzats:	Maxwell (M2B)	A24M
T1	Teoria:	Gibbs	A12G
(18:00)	Problemes:	Helmholtz	A12G
	Problemes	Bohr (T1A)	A12G
(19:00)	tutoritzats:	Einstein (T1B)	A24M

A. Subsection 1

This is the first subsection.

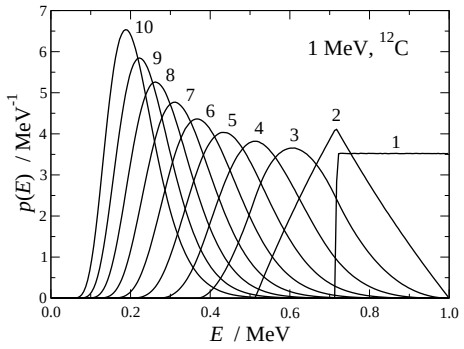


FIG. 2: A sample figure made smaller by changing the width parameter.

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B. Subsection 2

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IV. CONCLUSIONS

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Acknowledgments

Be sure to thank your advisor, your colleagues, and granting agencies (e.g. parents, etc...) as well.

[1] Helmut Kopka and Patrick W. Daly, *A Guide to LaTeX: Document Preparation for Beginners and Advanced Users*, 3rd. ed. (Addison-Wesley, Reading, MA, 1999).
[2] It is necessary to process a file twice to get the counters correct.
[3] J.D. Jackson, *Classical Electrodynamics*, (John-Wiley & Sons, New York 1975, 2nd. ed.).
[4] Hinshaw, G. et al. (WMAP Collaboration). "Five-Year Wilkinson Microwave Anisotropy Probe Observations: Data Processing, Sky Maps, and Basic Results". The Astrophysical Journal Supplement **180**: 225–245 (2009).
[5] Weinberg, S.. "High-Energy Behavior in Quantum Field Theory". Phys. Rev. **118**: 838–849 (1960).
[6] This is a footnote.

Preparació del manuscript del TFG amb LaTeX

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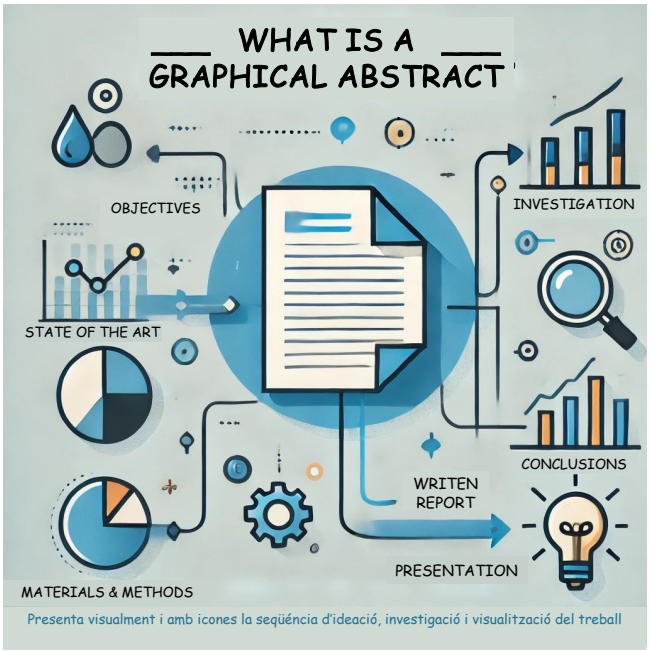
Resum: En aquesta sisena pàgina cal incloure la informació bàsica en la llengua oficial de la UB, que és el català. Concretament, és necessari el títol, el resum del contingut i les paraules clau. Es convida als estudiants a afegir-hi (opcional, però sempre dins de la sisena pàgina) un resum visual (Graphical Abstract), és a dir un resum únic, concís i visual de les principals conclusions del treball. Podria ser la figura del resultat final o, millor encara, una figura especialment dissenyada per a aquest propòsit, que mostri d'un sol cop d'ull el contingut del TFG. **Paraules clau:** Fins a 6 paraules que relacionin el treball amb conceptes estudiats al grau de Física.
ODSs: Aquest TFG està relacionat amb els Objectius de Desenvolupament Sostenible (SDGs)

Objectius de Desenvolupament Sostenible (ODSs o SDGs)

1. Fi de la es desigualtats		10. Reducció de les desigualtats	
2. Fam zero		11. Ciutats i comunitats sostenibles	
3. Salut i benestar		12. Consum i producció responsables	
4. Educació de qualitat	X	13. Acció climàtica	X
5. Igualtat de gènere		14. Vida submarina	
6. Aigua neta i sanejament		15. Vida terrestre	
7. Energia neta i sostenible	X	16. Pau, justícia i institucions sòlides	
8. Treball digne i creixement econòmic		17. Aliança pels objectius	
9. Indústria, innovació, infraestructures			

Es pot incloure un breu paràgraf subratllant la relació entre aquest TFG i els ODSs i les seves fites específiques (disponibles a https://www.diba.cat/web/ods/coneix-els-ods i a https://sdgs.un.org/goals). Per exemple: "El contingut d'aquest TFG, part d'un grau universitari de Física, es relaciona amb l'ODS 4, i en particular amb la fita 4.4, ja que contribueix a l'educació a nivell universitari. També es pot relacionar amb l'ODS 7, fita 7.2, perquè s'estudia energies renovables, i amb l'ODS 13, fita 13.4, perquè promou l'educació, mitigació, adaptació i reducció de l'impacte del canvi climàtic."

GRAPHICAL ABSTRACT (OPTIONAL)



Appendix A: SUPPLEMENTARY MATERIAL (OPTIONAL)

Here, you may add supplementary materials that are relevant but not essential if needed. These can include data from databases, programming code, figures, tables, equations, etc., in any format. Note that these materials will not be evaluated by the jury and may not even be viewed. For example, the following equation,

$$\vec{F} = m\vec{a} \tag{A1}$$

is important in physics.